

Pradyumna Kumar

Atlanta, GA (Open to relocate) | +1-943-224-6350 | pradyumna.ka@gmail.com | [LinkedIn](#) | [GitHub](#) | [Portfolio](#)

SUMMARY

Data Scientist and Machine Learning Engineer with 4 years of production experience translating complex data into measurable business outcomes. Developed production ML systems across recommendation, NLP, and predictive modeling – a \$3M/year cost reduction, 60% engagement lift, and 50% operational savings at a high-growth SaaS company. Strong foundation in statistical modeling, experimentation, and stakeholder communication, with published research applying AI/ML to healthcare and biotech.

SKILLS

Python, SQL (Postgres, MySQL), NoSQL (MongoDB), Bash, Java, Statistical Modeling, Feature Engineering, A/B Testing, Deep Learning, NLP, LLMs (GPT, Claude, Gemini), SLMs (Llama, Qwen), RAG, Agentic AI, Fine-tuning (LoRA, GRPO), Pandas, NumPy, Matplotlib, Scikit-learn, XGBoost, PyTorch, ETL, Databricks, Snowflake, AWS (S3, Redshift, SageMaker), GCP, Docker, MLflow, Excel, Tableau, Looker Studio, Git, Bitbucket, Jenkins.

EXPERIENCE

Machine Learning Researcher

May 2025 – Present

Kennesaw State University (Part-time)

Atlanta, GA

- Designed and evaluated a privacy-preserving Alzheimer's diagnostic pipeline by fine-tuning a 1B-parameter Small Language Model (SLM) on clinical speech transcripts, achieving 70% F1 on held-out validation data.
- Research on adversarial robustness of neural and pretrained vision models using Python and PyTorch, uncovering vulnerability patterns that reduce model confidence by up to 30% and guiding new mitigation strategies.
- Published research using ML, GPU-accelerated simulations, and statistical visualizations (seaborn, ggplot2) to model binding mechanisms, achieving 10x speedup over CPU baselines and 5 novel targets for wet lab validation.

Data Scientist - II

Feb 2021 – Aug. 2024

BetterPlace – Enterprise SaaS, Frontline Workforce Management

Bengaluru, India

- Replaced a \$3M/year third-party dependency by engineering a background verification system. Indexed court records into Elasticsearch, fine-tuned RoBERTa for name matching, and autoscaled on Kubernetes with KEDA.
- Built a job recommendation pipeline for 2M+ users, drove feature engineering, LightGBM modeling, and A/B testing. Deployed via FastAPI, Jenkins, and Kubernetes (EKS), increasing user engagement by 60%.
- Designed and delivered an AI-powered skill assessment agent using GPT-4 and Azure Speech-to-Text for automated fraud detection, reducing manual evaluation time and increasing candidate conversion through stakeholder demos.
- Built and owned batch and streaming ETL pipelines (Airflow, Kafka) that automated org-level data workflows, cutting operational costs by 50% and improving downstream data reliability across BigQuery and Redshift.

Decision Scientist Intern

Sep. 2020 – Dec. 2020

Mu Sigma Inc.

Bengaluru, India

- Analyzed retail footfall data using Python and Tableau to uncover traffic patterns that informed store layout decisions; built an ARIMA forecasting model for time-series prediction, reducing monitoring response time.

PROJECTS

ModelGate – Contract-Aware AI Routing | LLM Fine-tune (LoRA), FastAPI, JavaScript | *3rd, KSU Hackathon '26*

- Built an intelligent routing system that ingests enterprise contracts, extracts compliance constraints, and routes LLM queries to the optimal model – reducing inference cost by 59% with under 5% accuracy tradeoff.

ClinicalRAG – Clinical Vector Search Pipeline | PySpark, RAG, FAISS, NLP | *KSU C-Day Fall 2025*

- Designed a retrieval-augmented pipeline over clinical datasets (MTSamples) using FAISS vector search and PySpark for distributed preprocessing, enabling semantic search over unstructured medical records.

EDUCATION

Kennesaw State University

Atlanta, GA

Master of Science in Computer Science

May 2026

Bangalore Institute of Technology

Bengaluru, India

Bachelor of Engineering in Computer Science